



Reinforcement learning applied to machine vision: state of the art

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Abstract

Reinforcement learning (RL) is gaining a foothold in artificial intelligence-based research academia. More and more applications are coming to fore where RL is being applied in a novel and successful manner. As the areas of application are diverse, one important area of industrial and research significance having a high scope is machine vision. In this survey paper, the basics of RL are discussed first in order to give the reader an overview of the technology. Subsequently, the important, novel and upcoming state-of-the-art applications are discussed in the field of RL. The research areas discussed include image segmentation, object detection, object tracking, robotic vision, autonomous driving and image classification/retrieval. Various state-of-the-art works are discussed and an impression of the potential and impact of RL in the field of machine vision is built.

Keywords Reinforcement learning · Machine vision · Applications

1 Introduction

The general motive of machine vision (MV) [17] is the understanding of high level content of digital images belonging to various tasks like recognition of objects, scenery, or events, object classification, discovery of geometrical structure and various relationships between different objects, etc. Alternately, MV performs modeling of the real world as sourced from image and video receptions using visual sensors by recognition of the environment and estimation of the features.

In real-world applications, this understanding of the high-level content of images which have been received from various visual sensors, is the first action for solving different tasks like detection of road or pedestrians in autonomous driving vehicles. Alternately, particular decision-making issues are solved by extracting information from surrounding environment from the acquired images. Hence, the efficacy of image understanding technology must be estimated from the efficacy of various decisions made from the resulting understanding of the environment. Decisions like these are in turn determined by previous action results of the system.

As an example, navigational decisions of a robot are made from estimated localization of the robot being an outcome of specific MV task namely vision based positional estimation.

Contemporarily, one of the important driving technologies for high-level understanding of the environment from acquired images is the use of machine learning methods for MV tasks like image registration, image segmentation, image classification/retrieval, three-dimensional image reconstruction, object detection and object tracking, etc. Tasks like these have been formulated as problems to be learnt, and machine learning (ML) is essential and ubiquitous for the pattern extraction or regularity extraction from the images. Hence, ML is now a important component of MV. Reinforcement learning (RL) [78] is an upcoming ML technology, wherein learning is carried out by interacting with the environment. It also allows taking account of decision results and future actions from solutions of the corresponding MV tasks.

Reinforcement learning (RL) [78] has become one of the important areas of research in artificial intelligence (AI) today. The domains to which it has been applied are manifold. As machines become intelligent, the need to control and supervise them is fading away. The industry and research community is finding new ways to make the machines and various application-oriented-processes automated. There is an active frontier of research dedicated to this area. One such technique which is inspired from nature and how the brain works is reinforcement Learning (RL). It enables a reward-based learning scheme where the machine/task/application learns by interacting with its environment in a closed loop

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fashion. As more and more areas come under RL, machine learning is blending into a new dimension with the help of RL. One important machine learning task having industrial and research significance is machine vision. Machine vision is all around us. From optical character recognition (OCR)-based document scanners to state-of-the-art self-driving vehicles. Due to the paramount significance and relevance of machine vision in these rapidly developing times, this paper attempts to bring to the notice of the global industrial and research/academic community, the cutting-edge breakthroughs in the field of RL-based machine vision. Herein, the RL basics are discussed to give the reader an overview of the technology. Next, significant and state-of-the-art applications are discussed in the field of RL-based machine vision. The applications discussed include image segmentation, object detection, object tracking, robotic vision, autonomous driving and image classification/retrieval. Various state-of-the-art works in these application sub-domains are discussed in a beneficial and satisfactory manner. Finally, the issues with RL and its future scope is discussed.

2 Reinforcement learning

Reinforcement learning (RL) is a decade old area with significant research [78]. Much work has been done in enforcing the mathematical background of the area and more techniques are introduced constantly with time, so much so that the deep learning [36,42,90] area of research has been applied and successfully merged into RL providing a large array of applications e.g., using Deep Q-Networks [23]. In this section, we turn toward the basic algorithms of RL giving a brief introduction to each. We start with defining the RL framework.

In reinforcement learning, the agent interacts with the environment which is modeled as a Markov decision process (MDP) [2], which in turn is represented by a six-tuple $(S, A, p_0(S), T, R, \gamma)$ where:

- S is a state set,
- A is an action set,
- $p_0(S)$ is the initial distribution of the states,
- $T(S_{t+1}|S_t, A_t)$ is the transition probability state S_t to S_{t+1} , where $S_t, S_{t+1} \in S$ with action $A_t \in A$,
- $R(r_{t+1}|S_t, A_t)$ is the reward probability $r_{t+1} \in R$ after execution of action A_t in state S_t ,
- $\gamma \in [0, 1)$ being the discount factor

Now, we turn to value-based RL methods.

2.1 Value-based methods

These methods proceed by learning the value function for every state/state-action pair for the optimal policy π . Let

the state value function of a certain policy π be denoted as $V_\pi(s)$, $\forall s \in S$. Let the state-action value function be denoted as $Q_\pi(s, a)$, $\forall s, a \in (S, A)$. For finding the value functions of the optimal policy π^* , the update function is

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [r_{t+1} + \gamma \max_a Q(s_t, a) - Q(s_t, a_t)], \quad (1)$$

$$V(s_t) = \max_a Q(s_t, a), \quad (2)$$

converging to the optimal $Q^*(s_t, a_t)$ and $V^*(s_t)$.

For learning the estimate $\hat{Q}(\cdot, \cdot|\omega)$ with parameter ω for the optimal value function Q^* by using temporal-difference techniques, the following gradient update is used

$$\omega_{t+1} \leftarrow \omega_t + \alpha [r_{t+1} + \gamma \max_a \hat{Q}(s_t, a; \omega) - \hat{Q}(s_t, a_t; \omega)] \nabla_\omega \hat{Q}(s_t, a_t; \omega). \quad (3)$$

Here, optimal policy can be found by a greedy technique upon the optimal value function for every state

$$\pi^*(s) = \arg \max_a Q^*(s, a). \quad (4)$$

This technique in which the state-action value function is learnt with bootstrapping is known as *Q-learning* [91]. Value-based methods are also called *off-policy* methods because they are trained by using samples which are not taken from the policy being learnt.

Now, we turn to policy-based RL methods.

2.2 Policy-based methods

These methods directly learn the parameterized policy function π_θ instead of learning an explicit policy function, with the function parameters being θ . Policy gradients have used the REINFORCE technique [92] having the update function

$$\theta_{t+1} \leftarrow \theta_t + \alpha G_t \frac{\nabla_\theta \pi(A_t|S_t, \theta_t)}{\pi(A_t|S_t, \theta_t)}. \quad (5)$$

Here α being the step-size and $G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots$ being the return. The generalized version of the REINFORCE technique proceeds by using a baseline $b(S_t)$ to minimize the update function variance

$$\theta_{t+1} \leftarrow \theta_t + \alpha (G_t - b(S_t)) \frac{\nabla_\theta \pi(A_t|S_t, \theta_t)}{\pi(A_t|S_t, \theta_t)}. \quad (6)$$

The above-mentioned baseline is an estimate of the state value which is learnt separately as a parameterized function having weights ω . However, when the state value function

uses bootstrapping for the update as in value-based methods, then the method is referred to as an actor-critic method.

Now, we give a brief introduction to actor-critic RL methods.

2.3 Actor-critic methods

These methods [12] are both policy-based and hybrid-based and learn directly the policy, i.e., actor, as well as the value function, i.e., critic [39]. Their update function is of the form:

$$\theta_{t+1} \leftarrow \theta_t + \alpha \left(G_t - \hat{V}(S_t) \right) \frac{\nabla_{\theta} \pi(A_t | S_t, \theta_t)}{\pi(A_t | S_t, \theta_t)} \quad (7)$$

or

$$\theta_{t+1} \leftarrow \theta_t + \alpha \left(R_{t+1} + \gamma \hat{V}(S_{t+1}) - \hat{V}(S_t) \right) \frac{\nabla_{\theta} \pi(A_t | S_t, \theta_t)}{\pi(A_t | S_t, \theta_t)} \quad (8)$$

Here, $\hat{V}(\cdot)$ serves as the optimal value function's parameterized estimate. The respective update function for $\hat{V}(\cdot)$ is quite similar to that of *Q-learning* [91]:

$$\hat{V}(S_t) \leftarrow \hat{V}(S_t) + \alpha \left[r_{t+1} + \gamma \hat{V}(S_{t+1}) - \hat{V}(S_t) \right]. \quad (9)$$

In case of learning the estimate $\hat{V}(\cdot | \omega)$ having parameters given by ω belonging to the optimal value function V^* with the help of temporal-difference methods, the following gradient update is used:

$$\omega_{t+1} \leftarrow \omega_t + \alpha \left[r_{t+1} + \gamma \hat{V}(S_{t+1}; \omega) - \hat{V}(S_t; \omega) \right] \nabla_{\omega} \hat{V}(S_t; \omega). \quad (10)$$

In the work by [52] wherein *Advantage Actor Critic* or **A2C** was proposed, the authors formulated an on-policy RL technique based on the actor-critic technique having many parallel actors which replaced the value estimate function with the advantage $A_{\pi}(a, s) = Q_{\pi}(a, s) - V_{\pi}(s)$. Another notable contribution in this regard is the *Actor Critic using Kronecker-Factored Trust Region* technique or **ACKTR** wherein [94] used the trust region optimization along with the Kronecker-factored approximation (KFAC) [50] as well as the actor-critic technique. In trust region optimization [71], the update is limited to certain learning rate $\eta_{\max}$. ACKTR is defined by an optimization with a defined constraint in which the policy should not become too distant for the update. Also, in it steps are computed using the direction of the natural gradient instead of the direction of gradient. Since computation of the second derivative requires more resources, so [94] use an approximation given by the KFAC.

Recently, multi-agent systems [58,60,75,105] have been used in conjunction with the actor-critic technique [67,103].

As mentioned previously, with the dawn of deep learning, a new era of machine learning thrived. As intense research followed the inception of deep learning, subsequently it was applied to RL also in a very successful manner spawning the area of deep RL Learning. Hence, deep Q-Networks were born and many successful techniques and applications were born. In the next section, we aim to give an introduction to deep Q-Networks (DQNs).

2.4 Deep Q network (DQN)

In their work, [53] used a deep neural network for the purpose of the optimal action-value function approximation:

$$Q^*(s, a) = \max_{\pi} E \left[\sum_{s=0}^{\infty} \gamma^s r_{t+s} | s_t = s, a_t = a, \pi \right] \quad (11)$$

The above expression gives the maximum reward sum r_t with discount factor γ for every time step t . This is achieved by the policy $\pi = P(a|s)$ after making the observation.

Before publishing this work, it was known that reinforcement learning methods became unstable or divergent for nonlinear functions, e.g., when a neural network was used for representing Q . Minh et al addressed this issue by using the following. They used a biologically inspired technique referred to as *experience replay*. They also used an iterative update rule which adjusted the action values Q toward the target values Q' , the latter being updated periodically hence leading to a reduction in the target correlations. The *Q-learning* iteration-based update function of the DQN used a loss function

$$L_i(\theta_i) = E_{(s,a,r,s') \sim U(D)} \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta_i^-) - Q(s, a; \theta_i) \right)^2 \right] \quad (12)$$

Here, γ being the discount factor in agent horizon, θ_i being the DQN parameters for the i^{th} iteration, and θ_i^- being the DQN parameters for computation of the targets for the i^{th} iteration. θ_i^- which are the target network parameters are updated alongside the DQN parameters θ_i every C steps while being kept fixed between the updates.

In order to perform experience replay, the agent's experiences e_t are stored which are represented by a tuple:

$$e_t = (s_t, a_t, r_t, s_{t+1}) \quad (13)$$

which consists of the observed state, s_t , in the period t , the reward received, r_t , by the agent in the period t , the action taken, a_t , in the period t , and the resulting state, s_{t+1} , in

period $t+1$. This stored dataset of the experiences of the agent at the period t consists of the past experiences.

$$D_t = [e_1, e_2, \dots, e_t] \quad (14)$$

During DQN learning, Q -learning updating is computed based on samples/mini-batches of the experience (s, a, r, s') which are drawn uniformly and randomly from the collection of the stored samples D_t .

The maximum operator used in DQNs, uses same values for both selection as well as evaluation of the actions, which leads to higher inclination toward selecting overestimated values. This results in an over-optimistic value estimation. [86] prevented this by decoupling the selected component from the evaluated component. This technique led to *Double Q-learning* and *Double-DQNs* or **DDQNs**. In double Q -learning, the two functions are learnt by assignment of each experience randomly for updating in such a way that there are two sets of weights, θ and θ' . For each update, one set of weights is used greedily for policy determination, while as the other set of weights is used for value determination. By decoupling the respective selection and evaluation components for Q -learning and afterward rewriting the target, we obtain:

$$Y_t^Q \equiv R_{t+1} + \gamma Q \left(S_{t+1}, \arg \max_a Q(S_{t+1}, a; \theta_t); \theta_t \right). \quad (15)$$

The double Q -learning algorithm error can be written as the following:

$$Y_t^{DoubleQ} \equiv R_{t+1} + \gamma Q \left(S_{t+1}, \arg \max_a Q(S_{t+1}, a; \theta_t); \theta'_t \right). \quad (16)$$

For a deeper understanding of DQN working, the readers may refer to [15]. For study of two recent works in this regard, the readers may refer to [5,9]. In [5], the parallel processing power of neural networks is further used with ensembles of DQNs.

3 Reinforcement learning in machine vision

In this section, we proceed to discuss the state-of-the-art applications of reinforcement learning in the field of machine vision.

3.1 Image segmentation using active reinforcement learning

Active learning (AL) is a well-established field used for image segmentation [24] in which proceeds with selection

of the samples containing maximum information for labeling purposes for the learning algorithm to perform better given less data vis-a-vis a non-selective approach which labels the entire data collection. [30] combined heuristic metrics that encouraged the diversity as well as representativeness of the labeled samples. Other techniques use unsupervised over-segmentation based on super-pixels [40,85] which depend highly on the super-pixel segmentation quality. Some techniques focus on foreground and background heuristic segmentation of medical images [98] with hand-crafted techniques. Works like [48,72,87] use cost effective techniques by using acquisition functions for cost labelling of the images or the image regions. In spite of this, the information is not always provided leading to restriction of the applicability of the techniques. In their work [6], Casanova et al have applied data-driven reinforcement learning-based approach to image segmentation using active learning. Their technique uses Q -learning [91] for tackling pool-based active learning using novel sets of actions, states and rewards. They have adapted deep Q -Networks formulation for allowing the technique to be feasible computationally. The overview of the technique is shown in Fig. 1.

In their work [97], the authors proposed an RL-based search technique for optimization of the training strategy for 3-D segmentation of medical images and validated the same effectively on several segmentation tasks. The technique also has potential in conventional machine learning (ML) problems. As an example, the heuristics of the target learning algorithm could be conveniently determined after optimization or search for a certain medical imaging application. The authors of [74] have used RL-based techniques for image segmentation in *positron emission tomography* or **PET**. A common segmentation technique is thresholding relying on the image intensity which is suitable in PET as the latter has low resolution and also high contrast. Large variation in intensity of the pathology under consideration and also the noise lead to difficulty in thresholding. In order address these issues, the authors have done PET image thresholding by using a Q -learning-based technique.

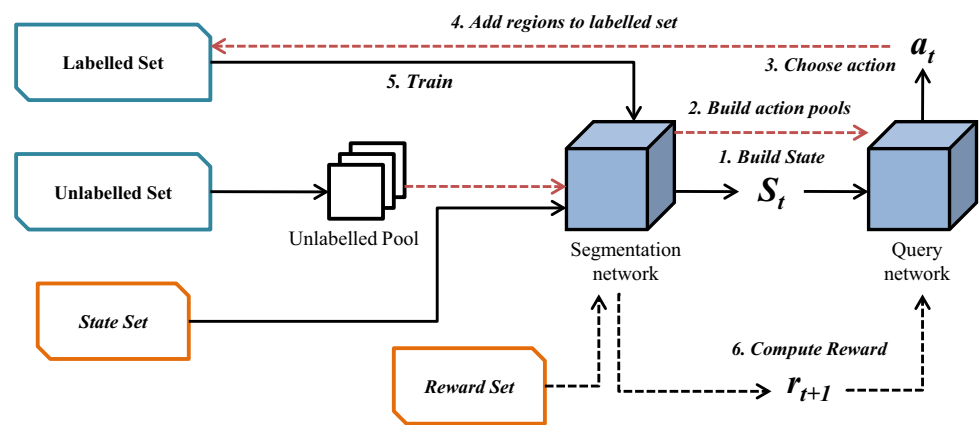
3.2 Object detection using reinforcement learning

Recently, reinforcement learning has been used for:

1. Replacement of classical object detectors, e.g., SSD [45] and Faster-RCNN [65]
2. Replacement of extensive proposal techniques in multi-stage object detection
3. Finding ROIs in large images where an object detector has to be run

Most of the techniques in the above categories use sequential learning policies. In category (1), in their works [4,51], the

Fig. 1 Overview of the RL-based image segmentation technique proposed in [6]



research groups proposed top-down models trained for object detection using Q-learning. In their work [4], the authors use an action set deforming the bounding boxes by translation and scaling, whereas in their work [51], the authors use an action set to locate bounding boxes in the image. RL methods for object detection generally fall in category (2). As an example, in their work [31], the authors propose an RL based technique which recursively divides up an image in a top-down fashion. The object proposals given by the agent are subsequently passed through the fast-RCNN network. Others use RL for finding object proposals sequentially to replace the first stage of a two-stage detector [19,63]. Works like [16,84], fall into category (3), wherein an RL agent undergoes training for examining down-sampled images and sequentially choosing region of interest (ROIs) for zooming into. In order to be efficient, works like [16] directly use detection in the fully downsampled image provided that the improvement gained by zooming in to an ROI is not high enough. Unlike the work done in [31], these techniques use a cost sensitive rewards for limiting the steps as has also been done in the works of [25,34]. They downsample the initially large image two times and focus on pedestrians which are reasonably pixelized objects. In their work [84], the authors consider a higher ratio for downsampling for the policy network for quite large remote sensing images wherein the box proposals are quite small. For the RL-based object detection techniques discussed above, the zoom-in ROIs given by them may contain more than one object instances, whereas in the techniques proposed in [19,31,51,63], search is undertaken for the bounding boxes surrounding a single object. The approach of [84] is different from other earlier predecessors as in that they proposed detection by learning sequential policies, whereas the former uses zoomed ROIs in a single iteration. This step has proved beneficial for making object detection parallelized using zoomed ROIs [93]. Overview of technique discussed in [84] is shown in Fig. 2.

3.3 Object tracking with reinforcement learning

Object tracking has emerged as an important area of research. IT involves real-time tracking of various objects in offline or online modes. Recently, deep RL has made rapid progress due to the application of deep learning to object tracking. Figure 3 shows the general architecture of a RL-based object tracking system.

In this regard, some works use sequential decision making techniques having learnt the policies with deep RL algorithms [77,100,102,107]. These techniques usually proceed by finding the best decision-making policy with RL by optimizing the tracking performance. In a related work, ActionDecision Network (ADNet) [100] uses a novel tracker which is controlled by sequential action set which is in turn obtained from deep RL stage. Also, ADNet combines supervised learning and RL during training. One more technique which relies on sequential action set is TrackDQN [99], which is a deep Q-Network-based approach for visual tracking. The proposed system has the decision-making ability to shift the target from its current state. TrackDQN uses a deep Q-Network for obtaining the action set for moving the Bounding Box (BBox) in the new frame. ADNet has 11 actions so that the BBox scale remains the same during tracking. Similarly, TrackDQN includes nine actions in its action set which include those for changes in aspect ratio. Recently RL-based techniques have been used in drone technology. For some recent works the readers may refer to those given in [1,73,104]. In a related novel work [20], the authors introduce a novel deep RL-based object tracking system for drone images by estimating an action series for finding the object location in the new frame. They introduce a new reward function for reduction in the actions for estimation of the location of the object in the new frame and also introduce a separate underlying network for use of LR images. They also introduce a new action set for better dealing with various complexities in the task.

Fig. 2 Overview of the RL-based object detection technique proposed in [84]. During training, high-resolution (HR) images are downsampled to low-resolution (LR) images. Then the agent uses the latter for sampling actions. The coarse/fine detectors use the LR/HR images to give bounding boxes. The bounding boxes are used for calculating reward with the help of labels from the HR images

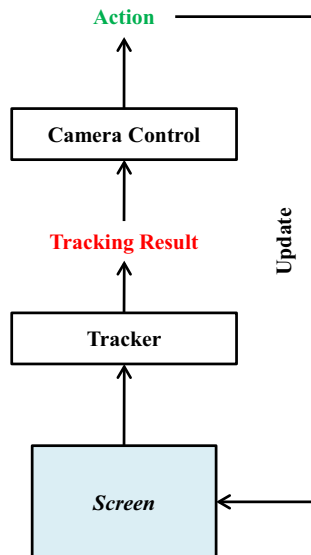
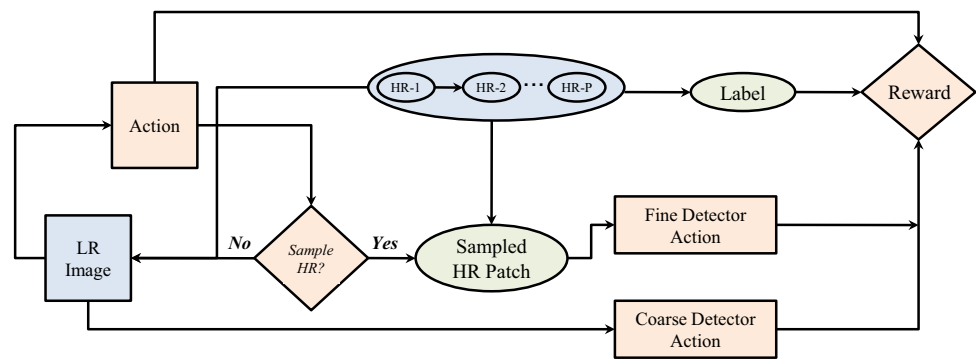


Fig. 3 Overall view of an RL-based object tracking system [47]

3.4 Robotic vision-based control with reinforcement learning

Manipulation by interacting with objects represents one of the biggest robotics problems. Interacting in an intelligent manner with unknown objects in various environments needs generalizable perception abilities, vision-based manipulation abilities, and dexterity. RL is an upcoming area in this regard however recent work on RL is used for handling individual skills, e.g., hitting a ball [62], opening the door [32,96], or throwing an object [18]. In order to meet the manipulation demands which are generalized in the real-world, scalable learning by using off-policy RL algorithms is one, e.g., for the task of grasping. Although grasping limits the problem of manipulation, it contains many of the biggest challenges in the area. For example, a robotic grasping system must have the ability to pick up unknown objects reliably and effectively while using actual sensors and actuators. Hence, grasping is a micro-environment for the bigger problem of robotic manipulation which provides challenges and is also

a practical application model problem for experimentation for generalizing robotic manipulation and varied interaction with objects. Most of the current work for robotic grasping is decomposed into the separate stages of sensing, planning, and action, i.e., perception of the scene and identification of suitable grasping locations, followed by path planning to the selected locations [49,54,83,101]. This methodology is not as per the grasping behaviors of humans and animals. In these behaviors, grasping is dynamic and very closely interleaves with the senses and also the control at each stage [3,7]. Also such a closed-loop system for grasping which is potentially trained for success on diverse tasks may also intelligently perform well on pre-grasping techniques, e.g., pushing and repositioning objects in order to provide an easier grasp. In spite of this, a big challenge for control of closed-loop grasping is that the sensor loop coupled with the motor(s) has to be closed for the images. This is quite difficult for effective utilization along with standard methods for optimal control using novel settings. In their work [33], Kalashnikov et al have investigated how off-policy deep RL can be used to acquire closed-loop dynamics in visual grasping tasks with the help of fully self-supervised data in order to generalize to unknown objects while testing the system. The low level, end-effector movement value has been predicted from the camera used, and the whole system has been trained with the help of real-world grasping. For more recent work in this regard, the reader may refer to the works given by [14,28,46,64,76,80].

3.5 Reinforcement learning for autonomous driving

Recently, deep RL has made outstanding achievements in high-dimensional space learning by using powerful deep neural networks [59]. In spite of this, deep Q-learning as well as policy gradient methods need interaction between the agent and the environment in order to obtain reward as well as feedback. Continuing with this issue, one obstacle is that it is not feasible to train an autonomous vehicle with RL in the real-world space because the vehicle can hurt objects in its surroundings if it makes an error. Performing RL with a

vehicle driving simulation software and subsequently transferring the learned model to the real-world space can lead to faster, cheaper training, and safer training than that done with a real vehicle. There are issues here as well. Sometimes models trained through simulation cannot generalize to real-world images [11,82]. In spite of this, work has been done in transfer-learning as well as domain-adaptation-learning in robotic for the purpose of simulation-to-real RL-based training [21,66,81]. These techniques proceed either by first training the model in the simulator and then fine-tuning the former in the real-world environment [66], or by learning to align the virtual images and real images by relating the two domains [82], or else by using randomly rendered simulation environments for training followed by testing in real-world environments [68,81]. The work of [66] proposes to use progressive network to transfer network weights from model trained on virtual data to the real environment and then fine-tune the model in a real setting.

For autonomous driving [38] can be applied to the following tasks: optimizing controller, planning of path and optimizing trajectory, planning motion and dynamic path, developing high-level policies for driving during complicated navigation, learning scenario-wise policy on highways, intersections, etc., learning reward with inverse RL using expert datasets for intention prediction of the pedestrians, vehicles, etc., and lastly learning policies for safety and risk estimation. Configuring the reward function for deep RL agents used for autonomous driving is an active research area. Some of the criteria for autonomous driving are distance travelled [13], speed [13,35,43], standstill [8], collisions [13,43], sidewalk infractions [13], in-line keeping, and maintenance of comfort and stability of the vehicle while avoiding extreme driving conditions [8,35], and obeying traffic rules [43]. Some autonomous driving tasks which require deep RL to learn a policy or behavior are discussed below:

3.5.1 Lane keeping

In their work [69], the authors propose a deep RL system for discrete actions as well as continuous actions using the TORCS autonomous driving simulation software [95]. However, it suffers from restrictions in terminating conditions and longer convergence in learning times. In their work [70], authors propose a system which learns discrete and continuous valued policies, respectively, using DQNs and deep deterministic actor critic (DDAC). The system follows the lane and maximizes average velocity. The system removes replay in DQNs leading to better convergence and better performance. Using a DDAC component in the system leads to action smoothing and better performance.

3.5.2 Lane changing

In their work [89], the authors use a Q-learning-based method for learning policy for the vehicle to do nothing, change lanes to left or right, accelerate, decelerate, etc. This technique is more robust in comparison with other contemporary techniques.

3.5.3 Ramp merging

In their work [88], the authors propose a recurrent architecture-based RL technique for modeling of long-term ramp merging policies for highway vehicles. State history is used for the task.

3.5.4 Overtaking

In their work [57], the authors propose a Q-Learning-based multi-goal RL policy for determining agent decisions based on information whether other vehicles interact with the agent. This system has better speed-tracking for lane keeping and overtaking without collisions.

3.5.5 Intersections

In their work [29], the authors propose a DQN-based systems for intersection negotiation. The action set defined by the authors enables better maneuvering around intersections with lesser visibility and restrictions.

3.5.6 Motion planning

In their work [37], the authors use an improved A* algorithm-based method for learning heuristic functions using deep neural networks for images of obstacle maps. The improvement using the technique is smoother control and better performance in comparison with DQNs.

3.6 Image classification/retrieval with reinforcement learning

A novel research area related to reinforcement learning is image classification. An initial body of work is presented in the papers of [10,55,56,106]. We proceed by discussing a related and novel recent work. Taking a cue from the maximum power point Tracking using in photovoltaic systems as given in the paper [27], the authors of [22] propose a strategy wherein they use a Q-Learning based technique for image classification. In this paper, a straightforward and robust hybrid image classifier is proposed wherein Q-Learning is used with 2 states and 2 or 3 actions. Other approaches use features extracted from convolutional neural networks

(CNNs) and use the same in the Q-state in addition to past history. This technique makes the other approaches complicated because of large number of states due to high-dimensional feature maps. Since their technique uses only 2 states it is simple and as such has much lesser number of parameters to optimize. Besides this, it also has a straightforward reward function. The proposed approach uses a novel action-set for image-processing. The proposed technique's performance is compared with others like ResNet50 [26], InceptionV3 [79], etc., on frequently used databases, e.g., ImageNet [41], *Cats and Dogs* Dataset [61], and Caltech-101 [44]. The proposed technique performs better in the experiments on the various datasets.

4 Major research algorithms, issues and trends

Though RL has seen significant progress in research, it has various issues.

The first major challenge for the research community is computational burden. More complex and more efficient RL techniques like DQN-based RL [53,86], demand a lot of computation resources. The DQNs are deep networks which require large datasets with a lot of images. Also, the DQNs have to be trained for thousands of epochs ensuring low error after extensive training.

The second major challenge of RL is in the ethical area [78]. Even the founders of RL admit [78] that there can be a time when a greedy algorithm like RL which bases its outcome on reward-based behavior may become 'too greedy' and harm its environment. If there is no check on the RL algorithm, it can 'loot' or 'plunder' the various resources available to it, due to its 'reward-consuming and maximizing' behavior. Thus, there is a need to put bounds and checks on RL-agents when it comes to exposing them to the real-world-based environment where various resources may have scanty guards and filters in place.

The third major challenge of RL is application of different techniques successfully to the modern-day problems faced in machine vision. It should also be mentioned that RL is showing progress and finding relevance in various areas like image segmentation, autonomous driving, image classification/retrieval, etc. Going into detail, in the area of RL-based image segmentation, Q-Learning has been used [6]. It will be interesting to see other state-of-the-art RL techniques being used in the same [74,91], especially for in vivo analysis. Again, in the area of RL-based object detection, Q-Learning has been used [4,51] and there is potential [51,84] for using other state-of-the-art techniques like DQNs [53,86] for the same. In this area, there is a need for higher intersection over union (IoU) thresholds for positive detections [84] while keeping the computational resources from getting strained,

and by using other recently developed RL techniques, the expectations can be met. In the area of RL-based object tracking, DQNs [53,86] continue to be the mainstay technique used. Issues like small object size, occlusion, object motion, etc., complicate tracking [20,73] and new techniques have to be found by using stronger and better RL techniques to overcome these issues. In the field of RL-based robotic vision, there is successful use of off-policy deep RL techniques [33]; however, the work so far is limited to certain frameworks and limited actions [32,33,62,96]. RL has the potential to be applied to a wider range of real-world applications. In RL-based autonomous driving, deep RL and policy gradient methods are used frequently [38]. However, as the area becomes more diversely applicable, on-road and off-road maneuvering as well as more control over variations in the route, terrain, etc., remain as issues [11,82] to be addressed. There is work going on also in this area in order to address these issues [38]. In RL-based image classification/retrieval, the work is very new [22]. Q-Learning has been applied, and there is scope for application of stronger and faster RL techniques [53,86] to the area. Bigger datasets [41] have to be used and the area has to be extended to real-time applications which are the expected future of this domain.

5 Conclusion

In this paper, the mathematical background of various reinforcement learning (RL)-based techniques was touched. Each RL theoretical domain was introduced along with relevant mathematical equations so that the reader can have a clear idea of the basics of RL. Next, the state-of-the-art applications of RL in machine vision were discussed one by one. These applications included image segmentation, object detection, object tracking, robotic vision, autonomous driving and image classification/retrieval. The state-of-the-art and other recent techniques were discussed along with major challenges and trends based on 107 papers from the research community.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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